

SCIENTIFIC REPORT · JOINTS SUBGROUP · BERTONE ET AL. 2026

Sustained through 26 weeks.

In joints, standard of care declined toward baseline; eCTM held pain relief and kept building performance.

Background

OBJECTIVE

Standard-of-care hemoderivatives (PRP, APS, IRAP) deliver soluble growth factors with a short biological half-life; prior controlled evidence in equine osteoarthritis showed benefit extending only to about 2 weeks. Whether a single dose of a biodegradable equine chorionic tissue matrix (eCTM) could sustain effect out to 26 weeks was the pre-specified durability question in the joints subgroup.

Methods

SUBGROUP

Joints subgroup of the 138-horse practice-based cohort. Blinded veterinarian scoring of lameness (AAEP), joint pain and swelling, and owner-reported performance return, at Weeks 0, 2, 4, 12, and 26. Mixed-effects models with treatment x time as fixed effect and horse as random.

PRIMARY ENDPOINT

Odds of return to or exceeding prior performance at Week 26, comparing eCTM vs. standard of care hemoderivatives in joints. Secondary: change in joint pain and performance scores across all visits, with attention to the Wk 12 to Wk 26 interval where standard-of-care effect was expected to decline.

MECHANISM (a priori)

Standard-of-care hemoderivatives release soluble factors that clear within days to weeks. eCTM is a biodegradable matrix that slowly presents fetal reparative proteins over months, providing a plausible basis for sustained effect beyond the standard-of-care window.

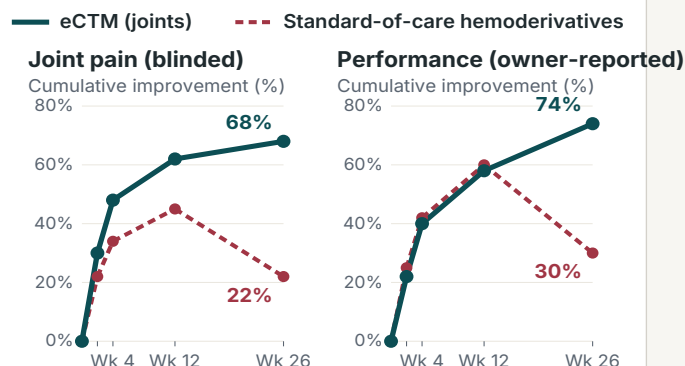
Results

RETURN TO PRIOR PERFORMANCE AT 26 WEEKS

2.72 : 1

Odds favoring eCTM over standard of care in joints at Wk 26 ($p < 0.006$).

FIG 1 · Joint trajectories through 26 weeks



Standard-of-care improvement peaked around Wk 12 then declined toward baseline by Wk 26. eCTM sustained pain relief and continued performance gains through Wk 26 ($p < 0.004$ vs standard of care at Wk 12; greater at Wk 26).

Interpretation

DURABILITY WHERE STANDARD-OF-CARE DECLINED

In joints, standard-of-care hemoderivatives produced a Wk 12 peak that declined toward baseline by Wk 26, consistent with the short pharmacology of soluble growth factors. A single dose of eCTM held pain relief and continued to build performance through Wk 26, with 2.72 : 1 odds of returning to prior performance favoring eCTM ($p < 0.006$). The pattern supports a longer-acting biologic effect and reduces the need for repeat dosing.

One dose. Sustained through 26 weeks.

In joints, eCTM held effect through 6 months while standard of care declined toward baseline between Wk 12 and Wk 26.

Equine Performance Labs